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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S6 (PT) (R, S) Examination May 2024 (2019 Scheme)

Course Code: EET302

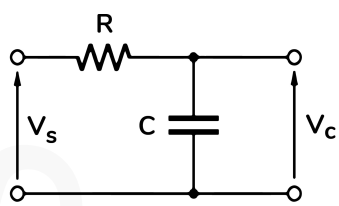
Course Name: LINEAR CONTROL SYSTEMS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. Marks

1 Define transfer function and find the transfer function of the given RC network with V_c as output and V_s as output. (3)



2 Give the characteristics of a lag compensator. (3)

3 Draw the unit step response of a standard second order under damped system and mark various time domain specifications. (3)

4 A unity feedback system has an open loop transfer function $G(s)H(s) = \frac{10(1+s)}{s((1+4s)((1+2s)}$. Determine the static error coefficients and steady state errors for unit step, unit ramp and unit parabolic inputs. (3)

5 What is root locus? Explain the information obtained from a root locus. (3)

6 List the characteristics of the PID controllers. (3)

7 Define resonant peak and resonant frequency. Draw the frequency response of a standard second order underdamped system and mark resonant peak and resonant frequency. (3)

8 Explain how the stability of a system is analysed using Bode plot. (3)

9 State Nyquist stability criterion. (3)

10 Explain the significance of Nichols chart. (3)

PART B

Answer one full question from each module, each carries 14 marks.

Module I

11	a)	Derive the transfer function of an armature controlled DC motor and represent the system in block diagram form.	(8)
	b)	With relevant characteristics explain the operation of a stepper motor	(6)
		OR	
12	a)	Determine the transfer function of a lead compensator using RC circuit components. List the characteristics of the lead compensators.	(8)
	b)	Explain the working principle of AC Tachogenerator.	(6)
		Module II	
13	a)	Derive the expressions for peak time and peak overshoot of an under damped second order system.	(8)
	b)	Determine the damping ratio, damped frequency of oscillation, rise time, peak time, peak overshoot and settling time for the system having closed loop transfer function, $\frac{C(s)}{R(s)} = \frac{80}{s^2+8s+160}$.	(6)
		OR	
14	a)	The forward path transfer function of a unity feedback control system is given by $G(s) = \frac{2}{s(s+3)}$. Determine the unit step response of the system.	(8)
	b)	Using Routh's stability criterion determine the stability of the given system whose characteristic equation is $s^6 + s^5 + 2s^4 + s^3 + 2s^2 + 5s + 6 = 0$.	(6)
		Module III	
15	a)	Determine the value of K for which the system has 0.5 damping ratio of the closed loop system with an open loop transfer function $G(s)H(s) = \frac{K}{s(s+2)(s+3)}$ using Root locus plot.	(10)
	b)	Compare the performance characteristics of PI and PD controllers.	(4)
		OR	
16	a)	Consider a unity feedback system with an open loop transfer function, $G(s) = \frac{K}{s(s+8)}$. Design a suitable compensator to meet the following specifications. (i) Percentage Peak overshoot = 9.5% (ii) Natural frequency of oscillation, ω_n =12 rad/sec (iii) Velocity error constant $K_v \geq 10$.	(10)

b) Explain the Ziegler Nichols method of tuning the PID controllers. (4)

Module IV

17 a) Sketch Bode Plot for the system with an open loop transfer function $G(s)H(s) = \frac{1}{s(1+s)(1+0.1s)}$. From Bode plot, determine gain margin and phase margin and assess the stability of the system. (10)

b) Explain transportation lag in control system? (4)

OR

18 a) Sketch Polar Plot for the given open loop transfer function $G(s)H(s) = \frac{1}{s(1+s)^2}$. Determine the gain margin and phase margin. (10)

b) Compare between minimum phase systems and non-minimum phase systems? (4)

Module V

19 Investigate the closed loop stability for the system having an open loop transfer function $G(s)H(s) = \frac{1}{s(1+s)(1+2s)}$, using Nyquist stability criterion. Also find gain margin and phase margin. (14)

OR

20 Design a phase lead compensator for a unity feedback system given by the open loop transfer function $G(s) = \frac{K}{s(s+1)}$ to meet the following specifications. (14)
(i) Phase margin of the system $\geq 45^0$ (ii) Steady state error for a unit ramp input $\leq \frac{1}{15}$. (iii) Gain cross over frequency of the system must be less than 7.5 rad/sec.

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